AI-Assisted Fetal Ultrasound Interpretation for African Resource-Limited Settings

Group 11 · Capstone Proposal · Oct 19, 2026

- Rhoda Mwende Kimweli
- Oscar Mokera
- FIDELIS NGUYO MURIITHI
- Rose Murugu
- Pauline Kaiuki
- Abraham Kiplimo

1. Problem and rationale

This capstone will build and stress-test a lightweight model that recognises fetal ultrasound planes and usable frames, using four public datasets that include real African and novice-operator data. Across much of sub-Saharan Africa, including rural Kenya, there are too few trained sonographers. Ultrasound devices are becoming cheaper, but interpreting what they capture remains the bottleneck. Complications such as malpresentation, growth restriction and placenta praevia then go undetected.

Blind-sweep protocols let midwives and nurses with about an hour of training acquire scans. What they still lack is support in judging whether a sweep holds a usable, diagnostic frame. Published work shows that models trained on European scans lose accuracy on African images, but recover quickly with small amounts of local data (Sendra-Balcells et al., 2023). Kenyan AIMIX data (more than 1,349 women; CORDIS) is not public. This study therefore uses open African data now and positions Kenyan validation as future work.

2. Objective and research questions

Main objective: to develop and evaluate a deep learning model that classifies fetal ultrasound planes and flags usable blind-sweep frames, robust to the devices, sites and operator skill found in African low-resource settings.

- **RQ1 (cross-region shift):** How well does a plane classifier pre-trained on European scans perform on African scans, before and after fine-tuning on a few African patients?
- **RQ2 (novice blind sweeps):** Can a model find frames usable for abdominal measurement in blind sweeps acquired by novice operators in Sierra Leone?

- **RQ3 (device shift):** How much does performance vary across ultrasound devices and sites, and which strategy best reduces that gap: transfer learning, domain adaptation or targeted augmentation?
- **RQ4 (deployment):** What recall threshold, model size and inference latency suit a screening-support tool on a low-cost device used by midwives?

3. Data sources

Four public, downloadable datasets cover every research question. Between them they hold about 17,000 labelled standard-plane images and 1,800 novice blind sweeps.

Dataset	Origin	Size	Labels we use	Answers	Access
ACOUSLIC-AI training set	3 public health units, Sierra Leone	300 cases × 6 sweeps (~840 frames each), 38 GB (we'll use the remote zip method-on python)	Per-pixel optimal / suboptimal / background abdominal plane; reference abdominal circumference (mm)	RQ2, RQ4	CC BY-NC-SA 4.0, Zenodo
African multi-centre fetal planes	Malawi, Egypt, Uganda, Ghana, Algeria	450 images, 120 patients	Abdomen, brain, femur, thorax (no thorax for Uganda or Ghana)	RQ1, RQ3	CC BY 4.0, Zenodo
FETAL_PLANES_DB	2 hospitals, Barcelona, Spain	~12,400 images, 6 classes	Abdomen, brain (3 sub-planes), femur, thorax, cervix, other; patient, machine and operator IDs	RQ1 (source domain), RQ3	Public, Zenodo
Multicentre-Fetal-Biometry	4 sites incl. UCL Hospital, 7 devices	4,513 images, 1,904 subjects	Head, abdomen, femur planes; device ID; biometry landmarks	RQ3 (device shift)	CC BY-NC-SA 4.0, figshare

Why these four. ACOUSLIC-AI is the only public set we found of blind sweeps by novices (1 hour of training) on a low-cost smartphone probe in Africa. It directly mirrors the midwife use case. The African planes set provides real-world evaluation across five countries; earlier work

showed fine-tuning with 4–6 African patients lifted AUC above 98% (Sendra-Balcells et al.). FETAL_PLANES_DB is the standard source domain for pre-training. The biometry set adds the device diversity the African data lacks (B2FIND record).

Not used as primary data: AIMIX Kenya (no public release; we will request access for future validation) and FPUS23 (phantom scans, not patients).

4. Methodology

The project follows CRISP-DM, adapted for medical imaging.

1. **Data preparation:** harmonise plane labels to a shared set (abdomen, brain/head, femur, thorax, other). Subsample ACOUSLIC-AI sweeps into frames labelled optimal, suboptimal or not usable. Resize, normalise intensity and remove on-screen text and device markers that could act as shortcuts.
2. **Exploratory analysis:** class balance by country, device and dataset; image intensity and artefact profiles; share of usable frames per sweep.
3. **Models:** a CNN trained from scratch on FETAL_PLANES_DB as a baseline, transfer learning with MobileNetV2 and EfficientNet-B0, and a few-shot fine-tune on small African subsets.
4. **Validation:** patient-grouped splits throughout to prevent leakage. Leave-one-country-out on the African set and leave-one-device-out on the biometry set.
5. **Deployment analysis:** model size, CPU inference latency, and the share of sweeps flagged for specialist review at the chosen threshold.

5. Evaluation, timeline and risks

Question	Metric	Target
RQ1	Macro F1 on African planes, before vs. after fine-tuning	Gap reported; ≥ 0.85 after fine-tuning
RQ2	Recall and precision for usable frames in ACOUSLIC-AI sweeps	Recall ≥ 0.80 for optimal frames
RQ3	Leave-one-device-out and leave-one-country-out F1	No device or country with recall < 0.50
RQ4	Model size, CPU latency, flag rate	< 20 MB; < 200 ms per frame; 15–30% of sweeps flagged

Timeline (16 Oct – 12 Nov 2026): Week 1, download and profile the data. Week 2, build the pipeline and baselines. Week 3, run transfer and adaptation experiments and the shift analysis. Week 4, deployment analysis, report and presentation.

Risks and mitigations

- *Compute:* ACOUSLIC-AI is 38 GB. We will subsample frames per sweep and work on cached tensors.
- *Label mismatch:* plane taxonomies differ across sets. We will map them to a shared set and report per-dataset results.
- *Ceiling effects:* fine-tuned plane accuracy may already be high. The primary findings are therefore the size of the shift and RQ2, not headline accuracy.
- *Non-commercial licences:* two datasets are CC BY-NC-SA. This suits academic use and will be stated in the README.

Limitations: no Kenyan data, no abnormality labels and no prospective clinical validation. Results are evidence for African low-resource settings in general, not approval for deployment.

Sources

- ACOUSLIC-AI challenge data and Zenodo record
- African fetal planes, Zenodo 7540448
- FETAL_PLANES_DB, Zenodo 3904280
- Multicentre-Fetal-Biometry (B2FIND)
- Sendra-Balcells et al., Scientific Reports 2023
- AIMIX reporting, CORDIS

Portions of this content were developed with AI assistance and reviewed by the project team.